

CPEN 416

Lecture 02: Single-qubit systems; introducing PennyLane

Tuesday 15 September 2026

Announcements

- Quiz 1 today
- Thursday class also on Zoom
- Practice assignment 1 available later this week

Recap from last time

Qubits are physical quantum systems with two **basis states**:

$$|0\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \quad |1\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

Arbitrary states are complex-valued linear combinations

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

where $|\alpha|^2 + |\beta|^2 = 1$ and $\alpha, \beta \in \mathbb{C}$.

Qubit states live in **Hilbert space**.

Recap from last time

Unitary matrices (gates/operations) modify a qubit's state.

A matrix U is unitary if

$$U U^\dagger = U^\dagger U = \mathbb{1} = I$$

They preserve normalization of state vectors (and angles between them - can prove on practice assignment 1).

We saw two examples:

$$X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \quad H = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$$

Learning outcomes

- 1 Compute measurement outcome probabilities of qubit states
- 2 Express quantum computations as quantum circuits, and as quantum functions in PennyLane
- 3 Represent the state of a single qubit on the Bloch sphere
- 4 Describe the behaviour of common single-qubit gates

Example: H

Perhaps the most important unitary in quantum computing is the Hadamard gate (H):

$$U U^\dagger = I$$

$$H^\dagger = H$$

$$H = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$$

$$|0\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

$$|1\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

This gate creates a *uniform superposition*:

$$H|0\rangle = \frac{1}{\sqrt{2}} (|0\rangle + |1\rangle) = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = |+\rangle$$

$$H|1\rangle = \frac{1}{\sqrt{2}} (|0\rangle - |1\rangle) = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix} = |-\rangle$$

$$H(H|0\rangle) = H|+\rangle = |0\rangle$$

$$H(H|1\rangle) = H|-\rangle = |1\rangle$$

Example: H

What if we apply it twice?

Example: Z

The gate

$$Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

does something a little different.

Apply to basis states:

$$Z |0\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix} = |0\rangle$$

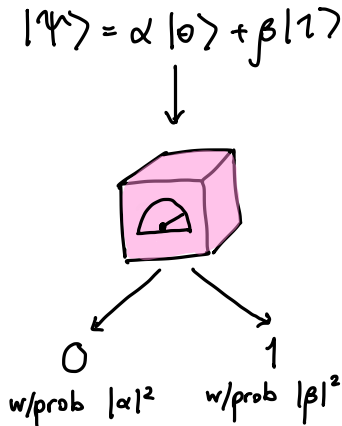
$$Z |1\rangle = \begin{pmatrix} 0 \\ -1 \end{pmatrix} = -|1\rangle$$

Measuring qubits

Manipulating qubits changes the amplitudes. Amplitudes determine probability of observing the qubit in $|0\rangle$ or $|1\rangle$ when we measure (it's **probabilistic**!).

Measuring gives us a single bit of information (0 or 1). To get more, we need to run the algorithm multiple times (multiple **shots**).

valid quantum states: $|\alpha|^2 + |\beta|^2 = 1$



Single-qubit gates: applying sequences of gates

$$H = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \quad Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

Exercise: A qubit starts in state $|0\rangle$. What is the probability of observing the qubit in state $|1\rangle$ after applying H , then Z ?

$$H|0\rangle = \frac{1}{\sqrt{2}} (|0\rangle + |1\rangle)$$

$$Z(H|0\rangle) = \frac{1}{\sqrt{2}} (Z|0\rangle + Z|1\rangle)$$

$$= \frac{1}{\sqrt{2}} (|0\rangle - |1\rangle) = \frac{1}{\sqrt{2}} |0\rangle - \frac{1}{\sqrt{2}} |1\rangle$$

$$|0\rangle \rightarrow \text{apply } H: \begin{matrix} \frac{1}{2} & |0\rangle \\ \frac{1}{2} & |1\rangle \end{matrix}$$

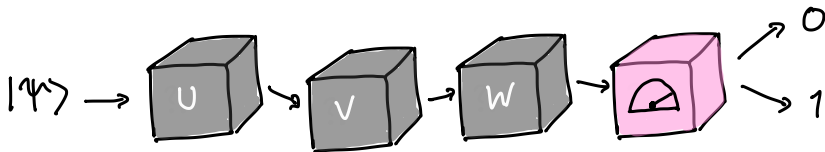
$$\rightarrow \text{apply } H \times 2: \begin{matrix} 1 & |0\rangle \\ 0 & |1\rangle \end{matrix}$$

$$\left| -\frac{1}{\sqrt{2}} \right|^2 = \frac{1}{2}$$

Quantum computing

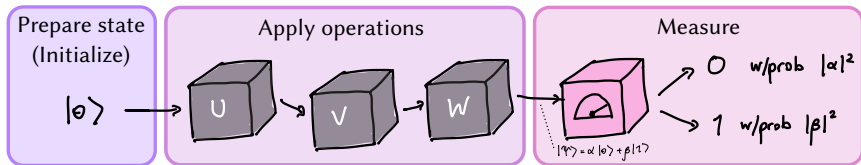
Quantum computing is the act of manipulating the state of qubits in a way that represents solution of a computational problem:

- 1 **Prepare** qubits in a **superposition**
- 2 Apply **operations** that **entangle** the qubits and manipulate the amplitudes
- 3 **Measure** qubits to extract an answer



Quantum circuits

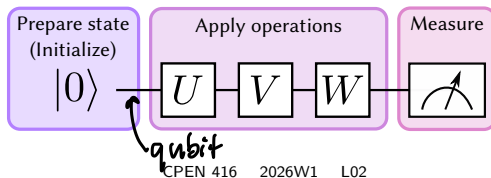
This representation is cumbersome:



So is the matrix representation:

$$Z H |0\rangle \quad Z Y \underbrace{W X U} |0\rangle = \dots$$

Alternative: quantum circuits!



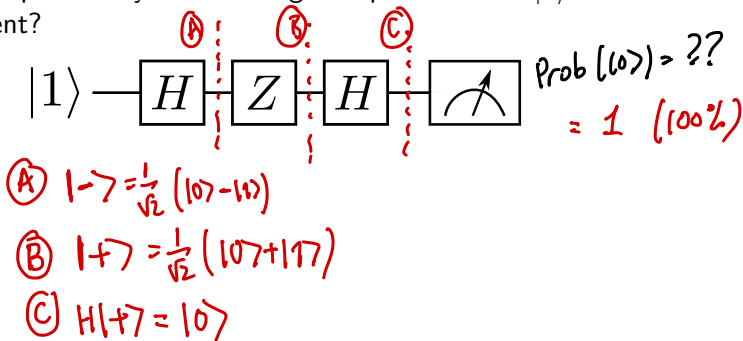
Quantum circuits

$$(H Z H) |1\rangle = |0\rangle$$

$$(H Z H) |0\rangle = |1\rangle$$

$$\underbrace{H Z H}_{\text{unitary}} = X!$$

Exercise: perform the quantum computation in the circuit below.
What is the probability of observing the qubit in state $|0\rangle$ after measurement?



Programming quantum computers

Everything we've done so far is just matrix-vector multiplication:

```
def ket_0():  
    return np.array([1.0, 0.0])  
  
def apply_ops(ops, state):  
    for op in ops:  
        state = np.dot(op, state)  
    return state  
  
def measure(state, num_samples=100):  
    prob_0 = state[0] * state[0].conj()  
    prob_1 = state[1] * state[1].conj()  
  
    samples = np.random.choice(  
        [0, 1], size=num_samples, p=[prob_0, prob_1]  
    )  
    return samples  
  
H = (1/np.sqrt(2)) * np.array([[1, 1], [1, -1]])  
X = np.array([[0, 1], [1, 0]])  
Z = np.array([[1, 0], [0, -1]])  
  
input_state = ket_0()  
output_state = apply_ops([H, X, Z], input_state)  
results = measure(output_state, num_samples=10)  
  
print(results)  
[1 0 0 1 0 0 0 1 1 0]
```

Sample NumPy

... better to use real quantum software instead.

PennyLane

PennyLane is a Python framework developed by **Xanadu** (a Toronto-based quantum startup).

```
import pennylane as qml

H = qml.Hamiltonian(...)

dev = qml.device('default.qubit', wires=2)

@qml.qnode(dev)
def quantum_circuit(params):
    qml.RY(params[0], wires=0)
    qml.RY(params[1], wires=1)
    qml.CNOT(wires=[0, 1])
    qml.RY(params[2], wires=0)
    return qml.expval(H)

quantum_circuit([0.1, 0.2, 0.3])
```

GitHub: <https://github.com/PennyLaneAI/PennyLane>

Documentation: <https://pennylane.readthedocs.io/en/stable/>

Demonstrations: <https://pennylane.ai/qml/demonstrations.html>

Discussion Forum: <https://discuss.pennylane.ai/>

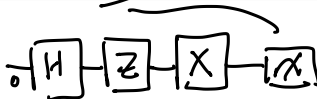
Quantum functions

We can express circuits as **quantum functions** in PennyLane. They are similar to normal Python functions, except they

- 1 apply one or more quantum operations to qubits (0-indexed)
- 2 must return a quantum measurement

```
import pennylane as qp

def my_quantum_function():
    qp.Hadamard(0) # Apply Hadamard gate to qubit 0
    qp.PauliZ(0) ↓ # Apply Pauli Z gate to qubit 0
    qp.PauliX(0) label # Apply Pauli X gate to qubit 0
    return qp.probs() # Return measurement probs
```



Devices

Quantum functions are executed on **devices**.

```
# Example 1: 5 qubits, analytic
dev = qp.device("default.qubit", wires=5)

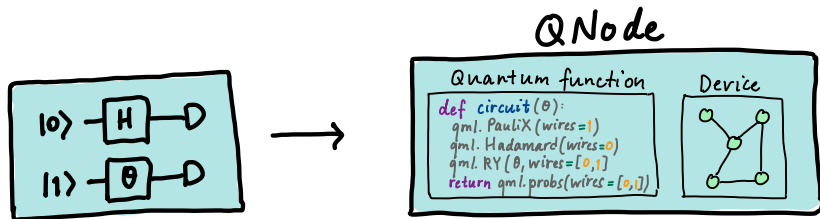
# Example 2: 1 qubit, probabilistic (100 measurements)
dev = qp.device("default.qubit", wires=1, shots=100)
```

Devices can be either *simulators*, or *actual quantum hardware*.

Qubits on a device are initialized in $|0\rangle$.

Quantum nodes

A **QNode** (quantum node) is an object that binds a quantum function to a device, and executes it.



```
# Create a QNode (version 1)
my_qnode = qp.QNode(my_quantum_function, dev)

# Execute the QNode
result = my_qnode()
```

Image: https://pennylane.ai/qml/glossary/quantum_node.html

Executing quantum computations with PennyLane

Exercise: perform the quantum computation below in PennyLane on a device with 1000 shots. What is the probability of observing the qubit in state $|0\rangle$ after measurement?



```
import pennylane as qp

dev = qp.device("default.qubit", wires=?, shots=?)

def my_quantum_function():
    # Your gates here
    return qp.probs()

my_qnode = qp.QNode(my_quantum_function, dev)
my_qnode()
```

Run locally, or try our interactive debugger, CircInspect: circinspect.ece.ubc.ca

You probably have some questions...

- ① Where's the state?
 - Inside the device
- ② How do gates get applied?
 - Operations are recorded onto a “tape”
 - The QNode constructs the tape when it is called
 - The tape is then executed on the device.

Single-qubit quantum states

So far we know the following 3 gates:

$$X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \quad H = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \quad Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

But, a general qubit state looks like

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle \quad |\alpha|^2 + |\beta|^2 = 1$$

What about the rest?

- 1 First, we'll explore a way to visualize arbitrary quantum states
- 2 Next we'll learn new unitary operations to create them with

Parametrization of qubit states

Exercise (part A): Consider the states

$$|\psi_1\rangle = \alpha|0\rangle + \beta|1\rangle, \quad |\psi_2\rangle = \alpha e^{i\phi}|0\rangle + \beta e^{i\phi}|1\rangle$$

global phase
 $|\psi_2\rangle = e^{i\phi}|\psi_1\rangle$
 $\sim |\psi_1\rangle$

How does $e^{i\phi}$ affect the measurement outcome probabilities of $|\psi_2\rangle$ compared to $|\psi_1\rangle$?

$|\psi_1\rangle$

$$\text{Prob}(|0\rangle) = |\alpha|^2$$

$$\text{Prob}(|1\rangle) = |\beta|^2$$

$|\psi_2\rangle$

$$\text{Prob}(|0\rangle) = |\alpha e^{i\phi}|^2 = |\alpha|^2 |e^{i\phi}|^2 = |\alpha|^2$$

$$\text{Prob}(|1\rangle) = |\beta|^2$$

Parametrization of qubit states

Exercise (part B): How many real numbers are required to *fully* specify a single-qubit state vector?

$$|\psi\rangle = \alpha |0\rangle + \beta |1\rangle \quad \alpha, \beta \in \mathbb{C}$$

2?

4?

$$\left. \begin{aligned} \alpha &= r_1 e^{i\phi_1} \\ \beta &= r_2 e^{i\phi_2} \end{aligned} \right\} 4$$

$$|\psi\rangle = r_1 e^{i\phi_1} |0\rangle + r_2 e^{i\phi_2} |1\rangle \quad r_1 / r_2 \Rightarrow \text{normalize}$$

$$= e^{i\phi_1} \left[r_1 |0\rangle + r_2 e^{i(\phi_2 - \phi_1)} |1\rangle \right] \quad \phi_1 / \phi_2 \Rightarrow \text{pull out global phase}$$

$$\sim \left[r_1 |0\rangle + r_2 e^{i\phi} |1\rangle \right]$$

$$= \cos \frac{\theta}{2} |0\rangle + \sin \frac{\theta}{2} \cdot e^{i\phi} |1\rangle$$

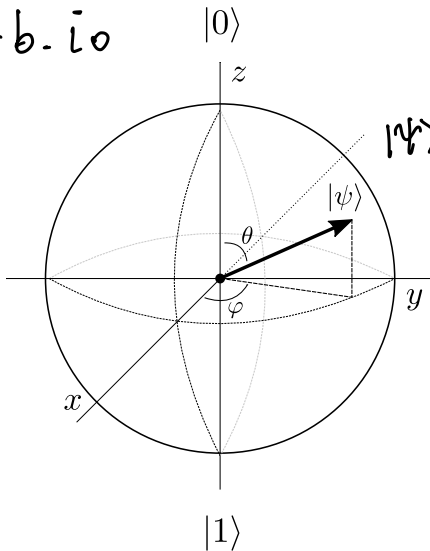
$$r_2 = \sqrt{1 - r_1^2}$$

$$\Rightarrow r_1 = \cos(\theta/2)$$

$$r_2 = \sin(\theta/2)$$

The Bloch sphere

bloch. kherb. io



$$|\psi\rangle = \cos\frac{\theta}{2} |0\rangle + e^{i\varphi} \sin\frac{\theta}{2} |1\rangle$$

Next time

Content: *Start with Bloch sphere + unitary ops*

- The theory of projective measurements
- Measuring in different bases

Action items:

- ① Work on problems in practice assignments
 - ② Explore the PennyLane docs and learn about the different kinds of operations and measurements
 - ③ Visualize combinations of operations on the Bloch sphere
- } later in week*

Recommended reading:

- Codebook modules IQC, SQ
- Nielsen & Chuang 4.2, 2.2.3, 2.2.5, 2.2.7